Commodity and Material Cost Index Review

H1 FY25 - input material rates and their effect on the open idea portfolio

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Reviewed by	Head of Cost Engineering
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Summary

Input material rates are the denominator of every value-engineering case. This review records the rates in force for H1 FY25 and re-tests the open idea portfolio against them. Hot-rolled coil has fallen 9.3 per cent over twenty-four months while aluminium has fallen 4.7 per cent from a higher base; the ratio between them has moved enough to change the ranking of eleven open ideas. Material-substitution cases built on rates older than twelve months should be re-tested before committee submission.

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1 Basis and scope

Rates in this review are landed costs at the Pune and Jamshedpur plants, inclusive of freight and handling and exclusive of recoverable taxes. They are averages across the qualified supplier panel and are weighted by FY24 purchase volume. They are appropriate for comparing design alternatives. They are not quotations and must not be used as a negotiating position.

1.1 Scope of materials

The schedule covers the material families that appear most often in value-engineering proposals: hot-rolled and cold-rolled steel, high-strength low-alloy grades, forging steels, aluminium alloys, engineering polymers and the principal coating systems. Fasteners are covered separately in CVOEM/VAVE/ES-STD/2023/004.

2 Rate movement

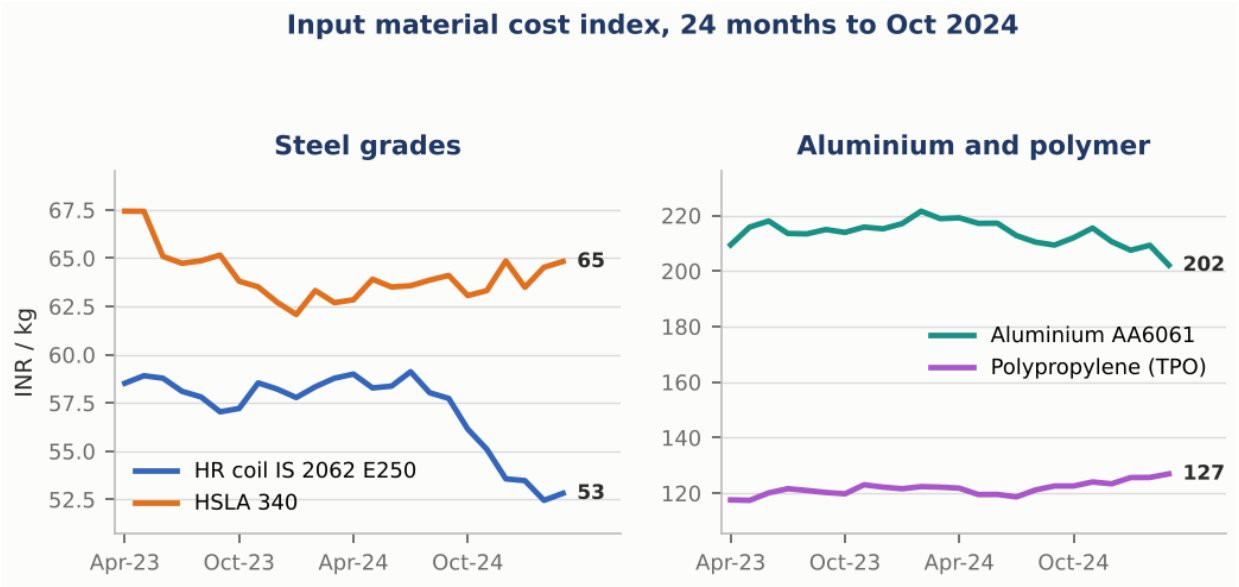


Figure 1 - Input material cost index over the twenty-four months to October 2024. Steel grades are shown separately from aluminium and polymer because the scales differ by an order of magnitude.

Material	Oct 2022	Oct 2023	Oct 2024	24-month change
HR coil, IS 2062 E250	58.4	57.8	53.0	-9.3%
CR coil, IS 513 D	64.2	63.1	59.4	-7.5%
HSLA 340	67.1	66.4	65.0	-3.1%
Forging steel, EN 8D	61.8	62.4	60.2	-2.6%
Micro-alloyed 38MnVS6	69.4	70.1	68.8	-0.9%
Aluminium AA6061-T6	212.0	206.4	202.0	-4.7%
Aluminium AA5754	198.6	194.2	191.8	-3.4%
Polypropylene, TPO grade	118.5	121.0	127.0	+7.2%
Glass-filled nylon PA66-GF30	268.0	281.4	294.6	+9.9%
Polycarbonate, lamp grade	341.0	352.8	368.2	+8.0%
Acrylic PMMA, lamp grade	246.5	251.0	258.4	+4.8%

All rates in INR per kilogram.

Finding 2.1. Steel has fallen and polymer has risen. The aluminium to HR-coil ratio has moved from 3.63 to 3.81 over twenty-four months, which makes steel-to-aluminium substitution marginally harder to justify on cost alone. Polymer substitution cases built on 2022 rates are materially overstated.

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3 Current rate schedule

3.1 Coating and surface treatment

Process	Unit	Rate (INR)	Note
Zinc plating, 8 um clear	per sq m	142	Chassis use discouraged
Zinc plating, 12 um yellow	per sq m	186	
Zinc-nickel, 8 um	per sq m	412	Meets 720 h requirement
Zinc flake, Dacromet class	per sq m	368	
E-coat	per sq m	224	
E-coat plus topcoat	per sq m	391	Highest performance
Powder coat	per sq m	268	

3.2 Process cost adders

Operation	Basis	Rate (INR)
Quench and temper, forged part	per kg	18.4
Controlled cooling from forging heat	per kg	4.2
MIG weld, structural	per metre	74.0
Robotic MIG weld, structural	per metre	41.5
Progressive press stroke	per stroke	2.8
Laser cut, 5 mm plate	per metre	118.0
Machining, milled face	per sq cm	6.4

Finding 3.1. Controlled cooling from forging heat costs INR 4.2 per kg against INR 18.4 per kg for a separate quench and temper cycle. On a 182 kg forged beam this is a difference of INR 2,585 in process cost before any material change, which is the largest single process delta in this schedule.

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4 Effect on the open idea portfolio

The open portfolio of 214 ideas was re-costed against October 2024 rates. Thirty-one ideas moved by more than INR 25 per vehicle, of which eleven crossed the committee threshold of INR 150 in one direction or the other.

Idea ref	Proposal	At submission	At Oct 2024	Movement
VC-23-0411	Steel to PP battery tray	INR 188	INR 142	Falls below
VC-23-0477	Steel to GF nylon radiator mount	INR 164	INR 118	Falls below
VC-24-0066	PC to PMMA lens, low impact	INR 131	INR 164	Rises above
VC-24-0119	HSLA upgrade, thinner crossmember	INR 142	INR 171	Rises above
VC-24-0188	Fabricated to pressed bracket	INR 147	INR 178	Rises above
VC-24-0233	Aluminium heat shield	INR 156	INR 149	Falls below
VC-24-0301	Micro-alloyed beam, direct cooled	INR 980	INR 1,240	Rises, already above

Finding 4.1. Two of the eleven ideas that crossed the threshold had already been closed as below benefit. They should be reopened rather than re-submitted, so that the original assessment and its validation notes remain attached.

4.1 Sensitivity of material substitution cases

Material substitution is the most rate-sensitive class of proposal because the benefit is the difference between two moving rates rather than a saving against a fixed baseline. A steel-to-polymer case assessed in October 2022 and unchanged since will overstate its benefit by between 11 and 18 per cent at current rates, depending on the polymer.

5 Guidance for idea submission

- Quote the rate basis date with every cost estimate. An estimate without a rate date cannot be re-tested.
- Material substitution cases older than twelve months must be re-costed before committee submission.
- Where a case depends on a rate within 10 per cent of the threshold, state the sensitivity explicitly rather than presenting a single figure.
- Process cost adders in section 3.2 should be used in preference to supplier quotations for comparing alternatives, so that proposals are ranked on a consistent basis.
- Ideas closed below the threshold are re-tested at each half-yearly review. Originators do not need to re-submit.

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